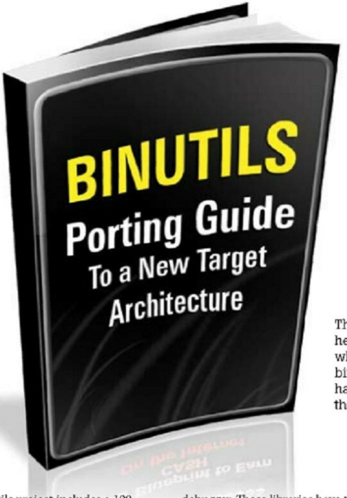

The article intends to help software engineers who want to port the binutils tools to a new hardware architecture for the first time.

Although the binutils project includes a 100-page guide to its internals, this article is aimed primarily at those wishing to develop/port binutils itself for the first time. The guide also suffers from the following limitations:

1. It tends to document at a detailed level. So individual functions are described well, but it is hard to get the big picture.
2. It is incomplete. Many of the most useful sections (for example, details of final relocation) are yet to be written.

Developers who require to port binutils to a new architecture are faced with discovering how binutils works by reading the source code and looking at how other architectures have been ported.

I, personally, went through that process when porting binutils to the Compact-RISC (a.k.a CR16) architecture. This article aims to capture the learning experience, with the intention of helping others, especially those looking to port binutils tools to a new target.

Binutils file organisation structure

The bulk of the binutils source code is in a small number of directories. Some components of binutils are libraries that are used internally as well as in other projects. For example, the BFD library is used in the GNU GDB debugger. These libraries have their own top-level directories. The main directories are shown in Figure 1.

Here, in brief, is some information on these directories:

- ***include*** contains the header files for information that straddles major components. For example, the main simulator interface header is here (*remote-sim.h*), because it links GDB (in the *gdb* directory) to the simulators (in the *sim* directory). Other headers, specific to a particular component, reside in the directory of that component.
- ***bfd*** contains the Binary File Descriptor library. This library contains code to handle specific binary file formats, such as ELF, COFF, SREC and so on. If a new object file type must be recognised, code to support it should be added here.
- ***opcodes*** contains the opcodes library. This has information on how to assemble and disassemble instructions.
- ***cpu*** contains source files for a utility called CGEN. This is a tool that can be used to automatically generate target-specific source files for the *opcodes* library, as well as for the SIM simulator used by GDB.
- ***binutils:*** Despite the name, this is not the main binutils directory. Rather, it is the directory for all of the binutils tools that do not have their own top-level source directory. This includes tools such as *objcopy,*